IERG4320/IEMS5726 — Chapter 2: Basic Statistical Test — Cheat Sheet

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Key Terminology

- **Population**: entire source of data
- **Sample**: a portion of the population
- **Variable**: any data item measurable or countable
- **Descriptive statistics**: characteristics of a population
- **Inferential statistics**: predictions for a population

Central Tendency

- **Mean**: $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
- **Median**: central value of ordered data
- **Mode**: most frequent value

Measures of Spread

- **Range**: $\max - \min$
- **Variance**: $\sigma^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$
- **Std Dev**: $\sigma = \sqrt{\sigma^2}$

Hypothesis Testing

Goal: Rule out chance (sampling error) as an explanation for observed results.

Steps

1. **State Hypotheses**
   - H_0 (null): no effect / no difference
   - H_a (alternative): effect exists
2. **Choose significance level** α (0.05, 0.01, or 0.1)
3. **Select test**: Z-test, t-test, χ^2, etc.
4. **Calculate test statistic** (e.g., z-score)
5. **Determine critical region**
6. **Decision**: Reject H_0 or fail to reject H_0

Critical Regions

One-tailed:

- Left-tail: reject H_0 if $z < -z_\alpha$
- Right-tail: reject H_0 if $z > z_\alpha$

Two-tailed: reject H_0 if $|z| > z_{\alpha/2}$

p-value

p-value = probability of observing results as extreme as (or more extreme than) the data, *assuming H_0 is true.*

Rule: If $p < \alpha$, reject H_0.

Example: Coin fairness test ($H_0 : p = 0.5$), 100 flips, 60 heads:
p-value $= 2\,P(X \geq 60) = 2(0.0287) = 0.0574 > 0.05$
$\Rightarrow$ Cannot reject H_0 at 95% significance.

Fisher's Exact Test

- For **2×2 contingency tables**
- Tests independence of two binary variables
- Computes **exact** probability (no approximation)

Contingency Table

	Group I	Group II	Total
Scene 1	a	b	$a+b$
Not Scene 1	c	d	$c+d$
Total	$a+c$	$b+d$	n

Example: Are dice fair?

Rolling 2 dice	Experiment	Theory
Sum ≥ 10	15	6
Sum < 10	57	30

```
import scipy.stats as stats
oddsratio, pvalue = stats.fisher_exact(
    [[15, 57], [6, 30]], "greater")
print(oddsratio, " ", pvalue)
# 1.3158   0.4051
```

$p = 0.405 \gg 0.05 \Rightarrow$ Cannot reject H_0.

Pros & Cons

Advantages: ideal for small samples; exact probability.
Disadvantages: only binary variables; slow for large n.

Chi-squared Test (χ^2 Test)

Tests whether observed frequencies differ significantly from expected frequencies.

Test Statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

- O_i: observed count of type i
- $E_i = Np_i$: expected count under H_0
- Degrees of freedom: $df = k - 1$

Chi-square Distribution

If $X_1, \ldots, X_k \sim N(0,1)$ are independent, then:

$$Y = X_1^2 + X_2^2 + \cdots + X_k^2 \sim \chi^2(k)$$

Properties: $E[Y] = k$, $\mathrm{Var}(Y) = 2k$.

Example: Biased die?

Die thrown 60 times: $O = [5, 8, 9, 8, 10, 20]$, $E_i = 10$.

i	1	2	3	4	5	6	Σ
O_i	5	8	9	8	10	20	60
E_i	10	10	10	10	10	10	60
$\frac{(O_i-E_i)^2}{E_i}$	2.5	0.4	0.1	0.4	0	10	**13.4**

```
from scipy.stats import chisquare
chi2, pvalue = chisquare([5,8,9,8,10,20])
print(chi2, " ", pvalue)
# 13.4   0.01991
```

Decision ($df = 5$, $\chi^2 = 13.4$):

- 95%: $13.4 > 11.070 \Rightarrow$ **Reject** H_0 (die is biased)
- 99%: $13.4 < 15.086 \Rightarrow$ Cannot reject H_0

Wilcoxon Rank-Sum Test

(Mann–Whitney U Test)

Non-parametric alternative to the two-sample t-test for **independent** samples.

Hypotheses & Assumptions

- H_0: $P(X > Y) = P(Y > X)$
- Observations are independent
- Under H_0, distributions of both populations are equal

Procedure

1. Rank all data from both groups (ascending); average ties
2. T = sum of ranks of one group
3. $U = T - \frac{n(n+1)}{2}$, $\quad n$ = group size
4. Normal approx: $\mu_U = \frac{n_1 n_2}{2}$, $\quad \sigma_U = \sqrt{\frac{n_1 n_2 (n_1+n_2+1)}{12}}$
5. $z = \frac{U - \mu_U}{\sigma_U}$, evaluate p-value

Example

X (15 values), Y (14 values):
$U_X = 272 - \frac{15\times16}{2} = 152, \quad U_Y = 58$
$\mu_U = 105, \quad \sigma_U = 56.125$
$z = \frac{152-105}{56.125} = 0.837, \quad p = 0.040$
Cannot reject H_0 at 99% level.

```
from scipy.stats import ranksums, mannwhitneyu
import numpy as np
X = np.array([3.99, 3.79, 3.60, 3.73, 3.21,
    3.60, 4.08, 3.61, 3.83, 3.31,
    4.13, 3.26, 3.54, 3.51, 2.71])
Y = np.array([3.18, 2.84, 2.90, 3.27, 3.85,
    3.52, 3.23, 2.76, 3.60, 3.75,
    3.59, 3.63, 2.38, 2.34])
U, pvalue = ranksums(X, Y)
print(U, pvalue)
# 2.0512  0.04024

statistic, pvalue = mannwhitneyu(X, Y)
print(statistic, pvalue)
# 152.0  0.04231
```

Wilcoxon Signed-Rank Test

Non-parametric alternative to the **paired** t-test for **dependent** samples.

Hypotheses & Assumptions

- H_0: two dependent samples come from same distribution
- Data are paired, each pair chosen randomly and independently

Procedure

1. Compute differences $d_i = x_i - y_i$
2. Exclude pairs where $d_i = 0$
3. Rank by $|d_i|$ (ascending); average ties
4. $W = \sum$ signed ranks
5. $\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{6}}$
6. Test stat: $t = \frac{W}{\sigma_W}$, evaluate with t-distribution ($df = n-1$)

Example

X	Y	$X-Y$	Rank
6.1	5.2	0.9	3.5
7.0	7.9	−0.9	3.5
8.2	3.9	4.3	10
7.6	4.7	2.9	7
6.5	5.3	1.2	5
8.4	5.4	3.0	8
6.9	4.2	2.7	6
6.7	6.1	0.6	2
7.4	3.8	3.6	9
5.8	6.3	−0.5	1

$W = 46, \quad \sigma_W = 19.62, \quad df = 9$
$P(|t_9| > 46/19.62) = 0.0438$
Cannot reject H_0 at 99% level.

```
from scipy.stats import wilcoxon
import numpy as np
X = np.array([6.1, 7.0, 8.2, 7.6, 6.5,
              8.4, 6.9, 6.7, 7.4, 5.8])
Y = np.array([5.2, 7.9, 3.9, 4.7, 5.3,
              5.4, 4.2, 6.1, 3.8, 6.3])
T, pvalue = wilcoxon(X, Y)
print(T, pvalue)
# 5.0  0.01953
```

Note: Python `wilcoxon()` uses T-statistic (min of positive/negative rank sums), which differs from hand-calculated W-statistic.

Quick Reference: Which Test?

Test	When to Use
Fisher's Exact	2×2 table, small sample, binary vars
χ^2 Test	Multi-category frequency comparison
Rank-Sum (U)	2 independent groups, non-normal data
Signed-Rank	Paired data, non-normal data

Core logic: Compute statistic → get p-value → if $p < \alpha$, reject H_0 → result is statistically significant.

Other Tests (For Reference)

- **t-test / Paired t-test**: compare two vars from same population
- **ANOVA**: extension of t-test for multiple groups
- **Correlation**: linear relationship between two vars
- **Regression**: predict unknown from known vars
- **Non-parametric tests**: no distribution assumptions